

Foundations of the Modern Web

An Engineering Perspective on Architecture, Protocols, and Infrastructure

The Digital Infrastructure

Distinguishing the backbone from the content

| Internet vs. World Wide Web

Understanding the distinction is fundamental to web architecture:

- ✓ **The Internet:** The global system of interconnected hardware—cables, routers, and networks using TCP/IP.
- ✓ **The WWW:** A software service running on top of the Internet, utilizing HTTP to link documents via hyperlinks.
- ✓ **Analogy:** The Internet is the highway; the Web is the vehicle moving across it.



| Client-Server Architecture



The Client

The "Consumer" of the data. Typically a web browser or mobile app that initiates a request for a specific resource.



The Request

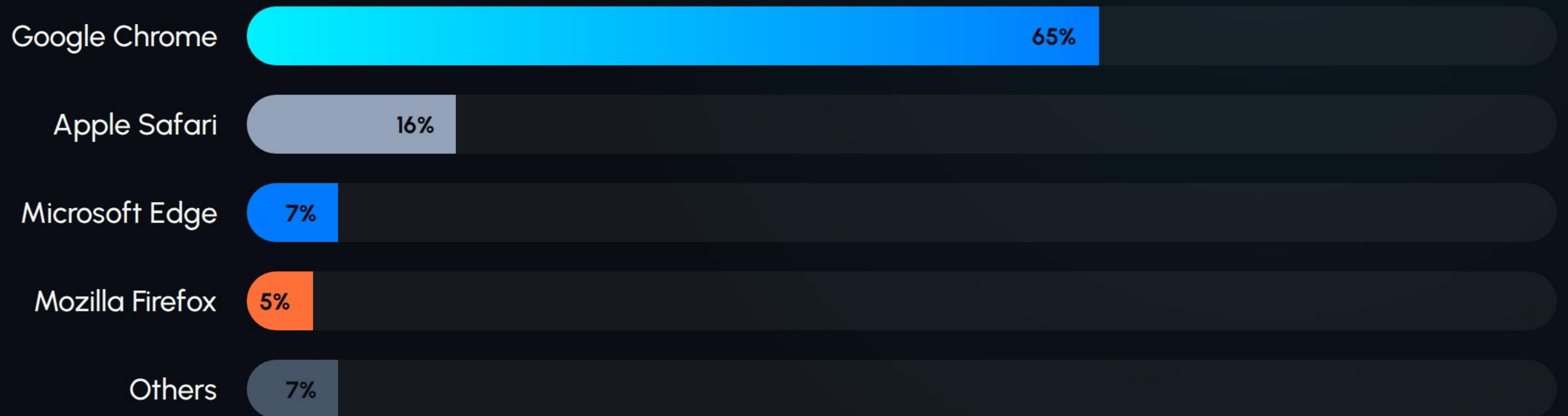
The communication channel. Requests travel via HTTP/HTTPS protocols over ISP networks to the target server.



The Server

The "Provider" of data. Hosts the physical files and listens for incoming requests to deliver HTML, CSS, or API data.

| 2025 Browser Market Share



Data highlights the continued dominance of the Chromium engine across desktop and mobile ecosystems.

| The Role of **Web Servers**

- ✓ **File Storage:** Servers store high-resolution assets, databases, and core application code.
- ✓ **Softwares:** Utilize industry standards like Nginx, Apache, or Microsoft IIS to manage traffic.
- ✓ **Request Handling:** Servers parse the incoming URL to retrieve the correct directory or data object.



| The Rise of Secure Browsing

95%
of Web Traffic is Secure

HTTPS: SSL/TLS Encryption

HyperText Transfer Protocol Secure (HTTPS) is no longer optional. It establishes an encrypted tunnel between the client and server.

- ✓ **Confidentiality:** Prevents eavesdropping.
- ✓ **Integrity:** Ensures data isn't altered in transit.
- ✓ **Authenticity:** Verifies the server's identity.

| Anatomy of a Web Request

- ✓ **DNS Lookup:** Browser translates *example.com* to an IP address (e.g., 93.184.216.34).
- ✓ **TCP Handshake:** Client and Server synchronize to prepare for data transfer.
- ✓ **HTTP GET Request:** The browser asks for the `index.html` file.
- ✓ **Server Processing:** The server locates the file and constructs a 200 OK response.
- ✓ **Content Rendering:** The browser parses HTML and requests CSS/JS files.
- ✓ **Visual Display:** Final assembly of the DOM to show the page to the user.

HTML: The Semantic Skeleton

HTML is the standard markup language used to structure web content. It defines the "meaning" of parts of a page.

- ✓ **Tags:** , ,
- ✓ **Attributes:** class, id, src
- ✓ **Validation:** Ensures accessibility and SEO.

```
/title>
```

```
ge: url("https://www.website.com/images/in
```

```
:no-repeat;
```

```
over;
```


| Standard Web Structure



The Header ()

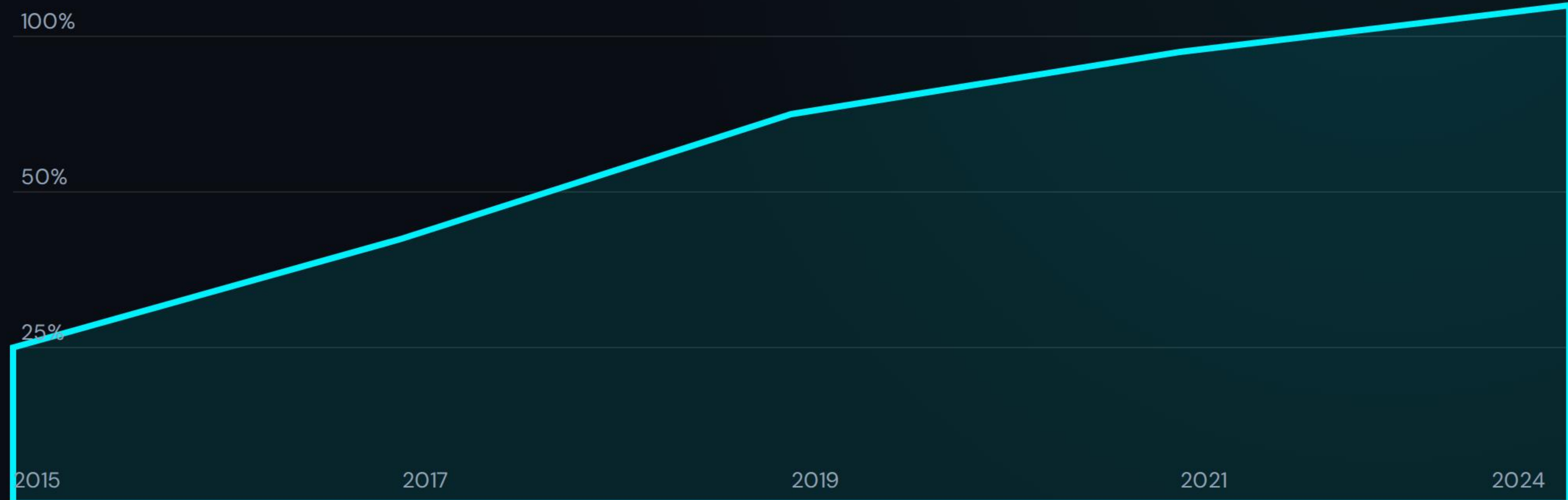
The "Brain" of the document. Contains invisible instructions: Metadata, character encoding, title tags, and links to external CSS stylesheets or Google Fonts.



The Body ()

The visible "Content". Every element seen by the user—text, images, buttons, and navigation menus—lives within this section of the code.

| HTTPS Adoption Trends



Significant growth in encryption driven by "HTTPS Everywhere" initiatives and browser security warnings.



Any Questions?

Deep dive into the architecture of the Internet and WWW completed.

Resources: [W3C Standard Documentation](#) | [MDN Web Docs](#) | [StatCounter](#)

| Image Sources



<https://media.istockphoto.com/id/2204318092/photo/physical-infrastructure-of-the-internet-visualization-inside-fiber-optic-cable-abstract-3d.jpg?s=612x612&w=0&k=20&c=tcOopwmCUDt-Y6wVPaHBOAyYoesJe47zXHQ862L9-rQ=>

Source: www.istockphoto.com



<https://static.vecteezy.com/system/resources/thumbnails/071/730/977/small/professional-of-a-modern-data-center-with-glowing-server-racks-and-network-cables-photo.jpeg>

Source: www.vecteezy.com



https://ik.imagekit.io/upgrad1/abroad-images/imageCompo/images/Type_backgroundOEPHGU.webp?pr=true

Source: www.upgrad.com
